The force tells us how motion changes

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Net force / ES: fuerza neta / PT: força resultante

Add signed forces along the same axis.

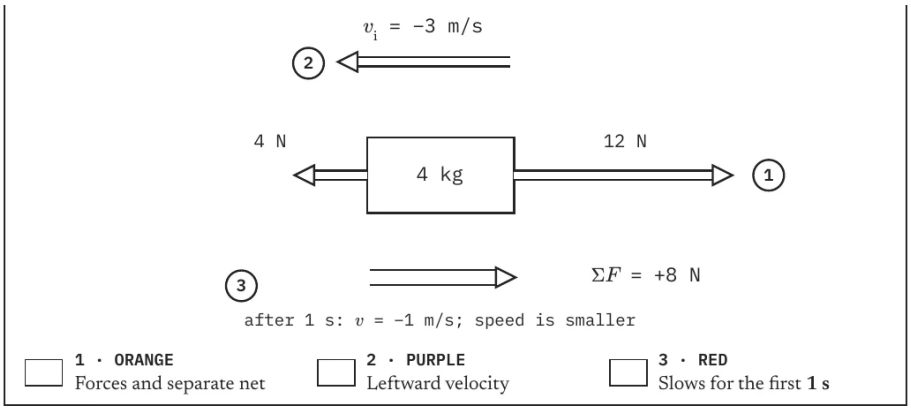
Also on your note page: Acceleration, Velocity, Speed trend.

B READ THE RELATIONSHIP

Add forces first. Then divide by mass.

As you read, find: force · sum · acceleration.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 4 kg cart has forces of 12 N right and 4 N left. Its initial velocity is –3 m/s . Find a and v after 1 s .

Given m = 4 kg F right = 12 N ; F left = 4 N v i = –3 m/s

Rule Sum forces first. Σ F x = ma x . v = v i + a x t .

Work Σ F x = 12 – 4 = +8 N a x = 8/4 = +2 m/s² v = –3 + 2(1)

Answer v = –1 m/s Still leftward; speed is now 1 m/s .

E SAVE FOR THE FINAL MEETING

The direction of XJ's net force tells us its acceleration direction. Its existing velocity must also be known to say whether it speeds up, slows down or turns.

Read the physics: Net force

SUPPORTED READING / read one part, then point to the evidence Name

01 PICTURE IT

Choose one object and a positive direction. Write forces to the right and left with opposite signs. Add before you divide by mass.

02 HOW IT WORKS

A net-force arrow summarizes the individual forces. Do not add that arrow again. Divide the signed total by mass. Acceleration points in the direction of the net force.

03 WHAT TO CHECK

Velocity can point the other way. A left-moving cart with a rightward acceleration slows at first.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 4 kg cart has forces of 12 N right and 4 N left. Its initial velocity is −3 m/s . Find a and v after 1 s . v = −1 m/s Still leftward; speed is now 1 m/s .

Model check: Add forces first. Then divide by mass.

05 CONNECT TO XJ

The direction of XJ's net force tells us its acceleration direction. Its existing velocity must also be known to say whether it speeds up, slows down or turns.

SAY IT BEFORE YOU WRITE

Which two steps give acceleration? I added ___ and ___ to get ___. Then I divided by ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Net force / ES: fuerza neta / PT: força resultante

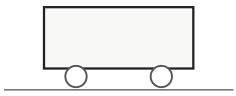
Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning	How can you tell?
To me, net force means …	A cart feels 12 N right and 4 N left. Which way is the overall push? How big is it? The two forces together give ___ N to the ___.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Draw an object with two opposing forces. Write the signed sum under the net arrow.

Labels you can use: force · sum · acceleration.



Tell a partner: I added ___ and ___ to get ___. Then I divided by ___.

On assignment: Net force

SECTION E / BACK / FINAL MEETING ONLY Name _____

A tracking model moves left. Its 4 kg body feels 12 N right and 4 N left. The status display has separate motion and force arrows.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Controls engineer**

Set the net-force and acceleration arrows. Tell the operator what happens to speed at first while velocity points left.

☐ **Replay analyst**

Annotate the slow-motion clip with one arrow for motion and another for acceleration. Explain why they can point opposite ways.

Word bank: force · sum · acceleration.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”

Editor’s check: Say what information is needed to predict when it reverses.